Perfect Storm Labs

Capabilities Statement

Company: Paul Swisher DBA FrankenBits **Division:** Perfect Storm Labs **Onwer/Operator:** Paul Swisher
Location: Washington State

Company Overview

FrankenBits' Perfect Storm Labs specializes in advanced acoustic detection systems and AI-powered monitoring platforms for defense, research, and environmental applications. Our patent-pending technologies provide real-time detection, classification, and correlation capabilities across multiple domains.

Core Competencies

Acoustic Detection & Classification

- Real-time acoustic signature detection and identification
- Multi-domain acoustic analysis (aircraft, atmospheric, seismic)
- Patent-pending acoustic detection methodologies
- Adaptive threshold triggering and anomaly detection
- Frequency spectrum analysis and pattern recognition

Artificial Intelligence & Machine Learning

- AeroMind AI - Aircraft classification and identification system
- Neural network-based pattern recognition
- Model deployment, monitoring, and drift detection
- Training pipeline automation
- Predictive analytics and correlation engines

Data Correlation & Validation

- Multi-source data fusion and consensus validation
- Real-time correlation of acoustic events with external data sources
- Cross-reference validation systems
- Weather and atmospheric propagation modeling
- Time-series data processing and analysis

Enterprise Software Architecture

- Microservices-based system design
- Service-oriented architecture (SOA)
- Triumvirate control plane for distributed systems
- PostgreSQL/TimescaleDB data architecture (16 databases, 122 tables)

- Real-time data pipeline engineering
- High-availability monitoring and alerting systems

Research & Development

- Acoustic meteorology and atmospheric monitoring
- Seismic event detection and precursor analysis
- Gravitational-acoustic coupling research
- Multi-sensor fusion and triangulation
- Advanced signal processing techniques

Patent Portfolio

Issued Patent Applications

63/928,429 - Acoustic Aircraft Detection (Filed: December 1, 2025)
Real-time acoustic signature identification and aircraft classification methodologies.

63/949,123 - Universal Triumvirate Control Plane (Filed: December 27, 2025)
Microservices orchestration and distributed system management architecture.

63/957,752 - Cosmic Gravitational Acoustic Resonance (Filed: January 2026)
Gravitational-acoustic detection methods with applications in seismic monitoring and cosmic event detection.

Technology Stack

Languages & Frameworks: Python, SQL, REST APIs
Databases: PostgreSQL 14+, TimescaleDB (time-series optimization)
Architecture: Microservices, Domain-Driven Design (DDD)
AI/ML: Neural networks, supervised learning, feature engineering
Data Processing: Real-time audio processing, signal filtering, FFT analysis
Integration: OpenSky Network, weather APIs, astronomical data sources

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Key Differentiators

✓ **Patent-protected innovations** - Three filed patents covering core detection methodologies
✓ **Multi-domain expertise** - Aircraft, atmospheric, seismic, and gravitational acoustic detection
✓ **Validated systems** - Real-world deployments with proven detection capabilities
✓ **Research-grade architecture** - Enterprise-scale data systems (16 databases, 122 tables)
✓ **AI-enhanced detection** - Machine learning classification with

continuous improvement
✓ **Multi-source validation** - Consensus-based correlation across independent data sources

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Applications

Defense & Security

- Aircraft detection and identification
- Airspace monitoring and violation detection
- Multi-sensor surveillance systems
- Real-time threat assessment

Environmental Monitoring

- Atmospheric acoustic analysis
- Weather system characterization
- Seismic event detection
- Environmental impact assessment

Research & Scientific

- Acoustic propagation modeling
- Gravitational-acoustic coupling studies
- Seismic precursor analysis
- Multi-domain correlation research

Past Performance

SeismoSentinel Deployment (2026)
- Multi-station seismic detection system - Validated 1,800-mile detection range - Real-time atmospheric acoustic-gravity wave monitoring

SentinelCore Platform (2025-2026)
- Real-time aircraft detection and correlation system - Multi-source data validation architecture - AeroMind AI classification engine deployment

Acoustic Meteorology Research (2025)
- 10+ hour weather prediction lead time validation - Atmospheric river detection and characterization - Novel acoustic weather analysis methodologies

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Contact Information

Perfect Storm Labs
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SAM.gov Registration: Active
CAGE Code: 17KK4
Primary NAICS Code: 541715 - Research and Development in the Physical, Engineering, and Life Sciences

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This capabilities statement contains high-level descriptions of patent-pending technologies. Detailed technical specifications and methodologies are available under NDA during proposal evaluation phases.

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